



## Background

The case starts at noon (12 UTC) of the 1st July 2006 and ends at 12 UTC the next day. During this period a stationary high pressure system resides over Scandinavia, North of Cabauw, resulting in cloud free conditions and a relatively constant Easterly geostrophic wind of approximately 8 m/s over Cabauw. The high pressure axis is tilted to the South over Cabauw resulting in a decreasing geostrophic wind with height. An airmass with dry air feature passes over the site at the start of the simulation period (12 UTC). From 00-03 UTC a small synoptical disturbance is advected over the site resulting in variations in temperature, humidity and wind.

### Boundary layer development

The case starts with a convective boundary layer height of approximately 1400 m. After sunset a turbulent stable boundary layer develops after sunset. Due to decoupling and inertial oscillation a low level jet develops during the night at 200 m height. At sunrise convective mixing gives a fast growing convective boundary layer up to 1900 m at noon.

### Prescribing the forcings

#### *Evaporation*

Evaporation from the grassland plays a dominant role in the surface energy budget, which results in relatively small sensible heat flux during day time (typically 100 W/m<sup>2</sup>). The modeller is asked to adjust model soil water such that the Bowen ratio is 0.33 at the start of the simulation.

#### *Roughness*

Roughness length for momentum is derived from wind observations. To arrive at a roughness length valid for a horizontal scale of a few kilometers use is made of wind gusts and standard deviation of the horizontal wind. This results in a regional roughness length of 0.15 m. Due to scattered obstacles like tree rows this is larger than the local roughness of 0.03 m for the grassland around the Cabauw tower.

The physical mechanism that extracts momentum from the flow by obstacles (pressure drag) is absent for temperature transport. Therefore a local estimate of roughness length for heat seems to be appropriate. Roughness length for heat is derived from observed surface radiation temperature of the local grassland, air temperature and sensible heat fluxes. A

typical value is 1.5 mm. When a model discriminates between these two roughnesses often a ratio of 10:1 is taken. Here the ratio is a factor of 10 larger due to obstacles on the regional scale.

#### *Initial conditions*

At the start of the simulation a dry air mass is advected over Cabauw. Using the corresponding radiosonde at that time to initialize the model will result in a too dry boundary layer and too low values for long wave incoming radiation. Since this first day time period is not of main interest for our evaluation we decided that this feature will not be forced to the model by prescribing advective tendencies but by increasing the humidity content on the basis of the previous and next 00 UTC soundings and tower observations. This results in a wetter and higher boundary layer at initial time than actual observed.

To avoid inertial oscillations at higher levels the wind is set equal to the geowind from heights of 2000 m upwards.

#### *Advection*

The disturbance at midnight can only be properly simulated when prescribing horizontal advective tendencies to the SCM's. The prescribed advective forcings are derived from 3D model simulations and a careful inspection of the observations.

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